

Technology Stack

Date	16 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	2 Marks

Technology Stack Details

AI BlogNest is a headless RESTful backend; the "client" during development and grading is an API tool (Postman / Thunder Client) rather than a browser UI, with a `Client` folder reserved for a future frontend.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Postman & the VS Code Thunder Client extension — used as the request/response client; the project's <code>Client/</code> folder is scaffolded for a future UI but out of scope for this cycle	Project scope for 15-29 Sept 2026 is a backend/API service; API clients are sufficient to exercise and grade every endpoint via the shared Thunder Client collection
2	Backend / Server-Side	Node.js + Express.js, with <code>cors</code> and <code>morgan</code> (dev request logging)	Non-blocking I/O suits the Gemini AI calls; minimal boilerplate for fast REST route development; whole team already proficient in JavaScript
3	Database / Data Storage	MongoDB with Mongoose ODM	Schema-flexible NoSQL store fits the evolving User/Blog/Comment documents; Mongoose adds field validation (required, unique, lowercase, minlength), defaults and <code>timestamps</code> on top of raw MongoDB
4	Cloud / Hosting / Deployment	API server run locally on port 5000 (<code>process.env.PORT 5000</code>) for the grading window; fully <code>.env</code> -driven configuration (<code>PORT</code> , <code>MONGO_URI</code> , Gemini key) so the API itself is cloud-deploy-ready	Keeps the grading environment simple and reproducible while leaving a clear path to a hosted MongoDB service and a cloud deploy without code changes
5	Version Control & CI/CD	Git & GitHub – github.com/sivraj0827/TNskills_AIblognest	Single source of truth for the 4-member team; commit history traces each Epic/Sprint
6	Third-Party APIs / Other Tools	Google Gemini AI (blog generation & summarization via <code>geminiService.js</code>), <code>dotenv</code> , <code>cors</code> , <code>morgan</code> , <code>nodemon</code> (dev)	Each library enables one specific, named feature of the solution — generative content, config isolation, cross-origin support, and request visibility during development